🚗 VFinance API — Rotas & Exemplos

Base URL: **http://10.0.0.147** | Direto na API: **http://10.0.0.147:3000** | Contrato: **0x3e299c…3305** | Chain: **Besu 1337**

> **Autenticação:** rotas marcadas com `AUTH` exigem um JWT. Gere em `POST /api/auth/token` e envie no header `Authorization: Bearer <token>`.

> **CORS:** liberado para qualquer origem (`Access-Control-Allow-Origin: *`), aplicado pela API . Preflight `OPTIONS` responde `204` .

> **Escritas on-chain** (registro, marca/modelo, status, hash, server-config) levam ~5–10s (mineração no Besu) e são assinadas pela chave do servidor (= `apiServerAddress`).

🔄 Fluxo E2E (ordem recomendada)

1. **Gera token** — `POST /api/auth/token` → guarda o JWT.
2. **Registra contrato** — `POST /api/contracts/` → retorna `regConId` , `metadataHash` , `txHash` .
3. **Consulta** — `GET /api/contracts/:regConId` → pega o `tokenId` e dados on/off-chain.
4. **Leituras on-chain** — por registry / chassi / hash / token, exists, brand/model, stats, total.
5. **Metadata off-chain** — store / get / update / delete.
6. **Escritas/admin** — registrar marca/modelo, atualizar status/hash, server-config, sync.

GET /health

Health check do serviço (exposto na rota dedicada `/health`).

CURL

```
curl http://10.0.0.147/health
```

200 OK

```
{ "status": "ok" }
```

POST /api/auth/token

Gera um JWT (role `admin`, validade 24h). Não requer body.

CURL

```
curl -X POST http://10.0.0.147/api/auth/token
```

200 OK


```
{
  "success": true,
  "message": "Token gerado com sucesso",
  "token":
"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJleHAiOjE3ODI0OTI3NjYsInJvbGUiOiJhZG1pbi
IsInVzZXJfaWQiOiJhZG1pbiJ9...."
}
```


GET /api/auth/validate AUTH

Valida o JWT do header e devolve as claims.

CURL

```
curl http://10.0.0.147/api/auth/validate \
  -H "Authorization: Bearer $TOKEN"
```

200 OK


```
{
  "valid": true, "success": true, "message": "Token válido",
  "user_id": "admin", "role": "admin", "exp": 1782492766
}
```

POST /api/contracts/ AUTH

Registra um contrato de financiamento (híbrido): grava dados primários on-chain (Besu) + metadados completos off-chain, ancorados por hash. **Obrigatórios:** `regConId`, `numeroContrato`, `dataContrato` e, dentro de `vehicleData`: `chassiVeiculo`, `placaVeiculo`, `marcaVeiculo`, `modeloVeiculo`, `valorTotalContrato` (> 0). Os demais campos vão para o metadata off-chain.

REQUEST BODY


```
{
  "regConId": "DOC-1782406308",
  "numeroContrato": "87686",
  "dataContrato": "2025-09-11",
  "vehicleData": {
    "regConId": "DOC-1782406308",
    "numeroContrato": "87686",
    "dataContrato": "2025-09-11",
    "nomeAgenteFinanceiro": "CREDORA INTERNA",
    "cnpjAgenteFinanceiro": "11204397000150",
    "cpfCnpjProprietario": "64687376487",
    "nomeProprietario": "ANDRE SANTOS",
    "chassiVeiculo": "983279794327897",
    "placaVeiculo": "PL87686",
    "marcaVeiculo": "FIAT",
    "modeloVeiculo": "ARGO",
    "modelName": "ARGO",
    "tipoRestricacaoContrato": "ALIENACAO FIDUCIARIA",
    "valorTotalContrato": "30000.00",
    "quantidadeParcelasContrato": "36",
    "taxaJurosMesContrato": "1.00",
    "veiculoZeroKm": true,
    "anoFabricacaoVeiculo": "2025",
    "anoModeloVeiculo": "2025"
  }
}
```


CURL

```
curl -X POST http://10.0.0.147/api/contracts/ \
  -H "Authorization: Bearer $TOKEN" \
  -H "Content-Type: application/json" \
  -d @contrato.json
```

201 CREATED

```
{
  "success": true,
  "message": "Contrato registrado com sucesso",
  "regConId": "DOC-1782406308",
  "metadataHash":
"854d3c0bc68726bb60f799310320787696303d4c87c0fbe7ee96f3fc98cff59a",
  "txHash": "0x4cbd7fcc2eb6d0a8fbb7a04b3fbf35f724892e2899f3d6567c531a3b993ecb33"
}
```


Erros: `400` campos faltando/inválidos · `InvalidInput` se `placaVeiculo` / `marcaVeiculo` / `modeloVeiculo` vazios · contrato/chassi duplicado.

GET /api/contracts/:regConId

Contrato híbrido completo: `data.onChain` (Besu) + `data.offChain` (todos os campos do contrato).

CURL

```
curl http://10.0.0.147/api/contracts/DOC-1782406308
```

200 OK (OFFCHAIN TRUNCADO)


```
{
  "success": true,
  "data": {
    "onChain": {
      "tokenId": "6",
      "regConId": "DOC-1782406308",
      "numeroContrato": "406308",
      "dataContrato": "1757548800",
      "metadataHash": "854d3c0b…cff59a",
      "timestamp": 1782406309,
      "registeredBy": "0xF08A60a634cbD8922B6e9c7aCFa391b402b80600",
      "active": true
    },
    "offChain": {
      "regConId": "DOC-1782406308", "chassiVeiculo": "983279794327897",
      "placaVeiculo": "PL87686", "marcaVeiculo": "FIAT", "modeloVeiculo":
"ARGO",
      "nomeProprietario": "ANDRE SANTOS", "cpfCnpjProprietario": "64687376487",
      "nomeAgenteFinanceiro": "CREDORA INTERNA", "valorTotalContrato":
"30000.00",
      "tipoRestricacaoContrato": "ALIENACAO FIDUCIARIA", "veiculoZeroKm": true
      /* + ~40 campos de contrato (taxas, datas, endereços, etc.) */
    }
  }
}
```


GET /api/contracts/chassis/:chassis

Mesmo payload híbrido, buscando pelo chassi.

CURL

```
curl http://10.0.0.147/api/contracts/chassis/983279794327897
```

GET /api/contracts/hash/:hash

Busca o contrato híbrido pelo `metadataHash`.

CURL

```
curl
http://10.0.0.147/api/contracts/hash/854d3c0bc68726bb60f799310320787696303d4c87c
0fbe7ee96f3fc98cff59a
```

GET /api/contracts/token/:tokenId

Busca o contrato híbrido pelo `tokenId` (numérico, on-chain).

CURL

```
curl http://10.0.0.147/api/contracts/token/6
```

GET /api/contracts/active?offset=0&limit=10

Lista paginada de contratos ativos (resumo). `limit` máx. 100.

CURL

```
curl "http://10.0.0.147/api/contracts/active?offset=0&limit=10"
```

200 OK


```
{
  "success": true,
  "data": {
    "contracts": [
      { "tokenId": "6", "regConId": "0x4778615d…149a0f", "numeroContrato":
"406308",
        "metadataHash": "854d3c0b…cff59a", "totalValue": "3000000",
        "brandName": "FIAT", "modelName": "ARGO", "active": true, "timestamp":
1782406309 }
      /* … */
    ],
    "total": 6, "offset": 0, "limit": 10
  }
}
```

GET /api/contracts/stats

Estatísticas agregadas (on-chain).

CURL

```
curl http://10.0.0.147/api/contracts/stats
```

200 OK


```
{ "success": true, "data": { "totalContracts": 6, "activeContracts": 6,
"contractVersion": "2.0.0" } }
```


GET /api/contracts/metadata-url/:hash · /api/contracts/metadata-url/registry/:registryId

Resolve a URL pública dos metadados (pelo hash ou pelo regConId).

CURL

```
curl http://10.0.0.147/api/contracts/metadata-url/registry/DOC-1782406308
```

200 OK


```
{ "success": true, "url": "http://localhost:8080/api/metadata/0x854d3c0b…cff59a"
}
```


POST /api/contracts/sync AUTH

Sincroniza dados on-chain → banco local (paginando 100/vez).

CURL

```
curl -X POST http://10.0.0.147/api/contracts/sync -H "Authorization: Bearer
$TOKEN"
```

200 OK


```
{ "success": true, "message": "Sincronização concluída com sucesso" }
```

GET /api/blockchain/version

Versão do contrato on-chain.

CURL

```
curl http://10.0.0.147/api/blockchain/version
```

200 OK


```
{ "success": true, "data": { "version": "2.0.0" } }
```


GET /api/blockchain/contracts/total

CURL

```
curl http://10.0.0.147/api/blockchain/contracts/total
```

200 OK


```
{ "success": true, "data": { "totalSupply": "6" } }
```

GET /api/blockchain/contract/registry/:registryId

Dados **puramente on-chain** (ContractRecord + VehicleCore). Note os campos em forma raw: `registryId` / `chassis` como hash/bytes32, `totalValue` sem casas decimais, `brandId` / `modelId` numéricos.

CURL

```
curl http://10.0.0.147/api/blockchain/contract/registry/DOC-1782406308
```

200 OK


```
{
  "success": true,
  "data": {
    "tokenId": "0",
    "contractRecord": {
      "registryId":
"0x4778615d5c7f20e4851988a18f84cdf3369e3a8694622e941f4ecf1014149a0f",
      "contractNumber": "406308", "contractDate": "1757548800",
      "metadataHash": "854d3c0b…cff59a", "timestamp": 1782406309,
      "registeredBy": "0xF08A60a634cbD8922B6e9c7aCFa391b402b80600", "active":
true
    },
    "vehicleCore": {
      "chassis": "983279794327897", "licensePlate": "PL87686",
      "totalValue": "3000000", "brandId": 4, "modelId": 1
    }
  }
}
```


GET /api/blockchain/contract/chassis/:chassis

Mesma estrutura on-chain, buscando pelo chassi.

CURL

```
curl http://10.0.0.147/api/blockchain/contract/chassis/983279794327897
```

GET /api/blockchain/contract/hash/:hash

CURL

```
curl http://10.0.0.147/api/blockchain/contract/hash/854d3c0b…cff59a
```

GET /api/blockchain/contract/token/:tokenId

CURL

```
curl http://10.0.0.147/api/blockchain/contract/token/6
```

GET /api/blockchain/contracts/active?offset=0&limit=10

Lista de tokenIds ativos direto da chain (`limit` máx. 100).

CURL

```
curl "http://10.0.0.147/api/blockchain/contracts/active?offset=0&limit=10"
```

GET /api/blockchain/contract/exists/:registryId

CURL

```
curl http://10.0.0.147/api/blockchain/contract/exists/DOC-1782406308
```

200 OK

```
{ "success": true, "data": { "registryId": "DOC-1782406308", "exists": true } }
```

GET /api/blockchain/hash/exists/:hash

CURL

```
curl http://10.0.0.147/api/blockchain/hash/exists/854d3c0b…cff59a
```

200 OK

```
{ "success": true, "data": { "exists": true } }
```

GET /api/blockchain/brand/:brandId

CURL

```
curl http://10.0.0.147/api/blockchain/brand/1
```

200 OK

```
{ "success": true, "data": { "brandId": 1, "brandName": "TOYOTA" } }
```

GET /api/blockchain/model/:modelId

CURL

```
curl http://10.0.0.147/api/blockchain/model/1
```

200 OK

```
{ "success": true, "data": { "modelId": 1, "modelName": "ARGO" } }
```

GET /api/blockchain/metadata-url/:hash · /api/blockchain/metadata-url/registry/:registryId

Resolve a URL de metadados direto da chain (`tokenURI`).

CURL

```
curl http://10.0.0.147/api/blockchain/metadata-url/registry/DOC-1782406308
```

POST /api/metadata/:hash AUTH

Armazena/atualiza os metadados (objeto `VehicleData`) sob um hash.

REQUEST BODY

```
{
  "regConId": "DOC-1782406308", "chassiVeiculo": "983279794327897",
  "marcaVeiculo": "FIAT", "modeloVeiculo": "ARGO",
  "valorTotalContrato": "30000.00", "veiculoZeroKm": true
}
```

CURL

```
curl -X POST http://10.0.0.147/api/metadata/0xabc123 \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d @metadata.json
```

201 CREATED

```
{ "success": true, "message": "Metadados armazenados com sucesso" }
```

GET /api/metadata/:hash

Recupera os metadados armazenados sob o hash.

CURL

```
curl http://10.0.0.147/api/metadata/0xabc123
```

200 OK (CAMPOS VAZIOS OMITIDOS)


```
{ "success": true, "data": { "regConId": "DOC-1782406308", "chassiVeiculo":
"983279794327897", "marcaVeiculo": "FIAT", "modeloVeiculo": "ARGO",
"valorTotalContrato": "30000.00", "veiculoZeroKm": true } }
```


PUT /api/metadata/:hash AUTH

Atualiza os metadados sob o hash (mesmo corpo `VehicleData`).

CURL

```
curl -X PUT http://10.0.0.147/api/metadata/0xabc123 \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"marcaVeiculo":"FIAT","modeloVeiculo":"ARGO HGT"}'
```

200 OK


```
{ "success": true }
```


DELETE /api/metadata/:hash AUTH

CURL

```
curl -X DELETE http://10.0.0.147/api/metadata/0xabc123 -H "Authorization: Bearer
$TOKEN"
```

200 OK


```
{ "success": true }
```

POST /api/blockchain/admin/brand AUTH

Registra uma marca on-chain (apenas owner). ~6s.

REQUEST BODY & CURL

```
curl -X POST http://10.0.0.147/api/blockchain/admin/brand \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"brandName":"VOLKSWAGEN"}'
```

200 OK

```
{ "success": true, "message": "Brand registered successfully", "txHash":
"0xcc0b5b25…fe9b24" }
```

POST /api/blockchain/admin/model AUTH

Registra um modelo on-chain. **Exige** `brandId` (uint) **e** `modelName`.

REQUEST BODY & CURL

```
curl -X POST http://10.0.0.147/api/blockchain/admin/model \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"brandId":1,"modelName":"COROLLA"}'
```

200 OK

```
{ "success": true, "message": "Model registered successfully", "txHash":
"0x80c4bd20…f40579" }
```

PUT /api/blockchain/contract/metadata AUTH

Atualiza o `metadataHash` de um token on-chain. O hash precisa ser único (bytes32, `0x` + 64 hex).

REQUEST BODY & CURL

```
curl -X PUT http://10.0.0.147/api/blockchain/contract/metadata \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"tokenId":"6","newMetadataHash":"0x1234…cdef"}'
```

200 OK

```
{ "success": true, "message": "Hash de metadados atualizado com sucesso",
"tokenId": "6", "txHash": "0xa552135e…04c6b" }
```

PUT /api/blockchain/contract/status AUTH

Ativa/desativa um contrato on-chain.

REQUEST BODY & CURL

```
curl -X PUT http://10.0.0.147/api/blockchain/contract/status \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d '{"tokenId":"6","active":true}'
```

200 OK

```
{ "success": true, "message": "Status do contrato atualizado para ativo",
"tokenId": "6", "active": true, "txHash": "0x6e0ba36a…1fd2b9f4" }
```

PUT /api/blockchain/admin/server-config AUTH

Atualiza a base URL de metadados e o endereço do servidor (apiServer) on-chain. **Campos:** `serverAddress`, `metadataBaseUrl`.

> ⚠ Trocar `serverAddress` para um endereço cuja chave a API não controla **bloqueia** futuras escritas. Mantenha `0xF08A60a634cbD8922B6e9c7aCFa391b402b80600` (chave atual do servidor).

REQUEST BODY & CURL

```
curl -X PUT http://10.0.0.147/api/blockchain/admin/server-config \
  -H "Authorization: Bearer $TOKEN" -H "Content-Type: application/json" \
  -d
'{"serverAddress":"0xF08A60a634cbD8922B6e9c7aCFa391b402b80600","metadataBaseUrl"
:"http://localhost:8080"}'
```

200 OK

```
{ "success": true, "message": "Server config updated successfully", "txHash":
"0x6e7c3a01…b8a1ea" }
```